Adversarial Fine-Tuning Improves NLI Robustness: An Empirical Study Using SNLI and ANLI

Anonymous Final Project Submission

Abstract

Pretrained language models perform well on natural language inference (NLI) benchmarks such as SNLI, but their success often reflects dataset artifacts rather than genuine reasoning ability. Prior work shows that NLI datasets contain statistical shortcuts including hypothesis-only cues, syntactic heuristics, and length-based correlations, causing models to fail on adversarial examples. In this work, we evaluate the robustness of the ELECTRA-small model trained on SNLI and progressively fine-tuned on the adversarial ANLI dataset (Rounds 1–3). Through analyses of accuracy, confusion matrices, curated reasoning categories, and length-based prediction behaviors, we find that ANLI fine-tuning substantially improves overall ANLI dev accuracy and entailment recognition, reduces harmful overreliance on predicting neutral, and mitigates strong length-driven biases present in the SNLI-only model. While correctly predicting contradiction for adversarial examples still remains a challenge, our qualitative and quantitative analyses indicate that ANLI fine-tuning teaches ELECTRA-small deeper semantic reasoning patterns that SNLI alone does not provide, and does so without substantially reducing SNLI performance.

1 Introduction

Pretrained language models achieve strong performance on natural language inference (NLI) benchmarks such as the Stanford Natural Language Inference (SNLI) dataset [1], but these gains often arise from dataset artifacts rather than robust inference. Prior work shows that NLI datasets contain shortcut features including hypothesis-only cues [2], syntactic heuristics [3], and hypothesis and premise length based correlations [4]. Consequently, models frequently achieve high in-domain accuracy while performing poorly on adversarial examples.

The Facebook Adversarial NLI (ANLI) dataset [5] was introduced specifically to expose weaknesses in NLI systems. ANLI is constructed through iterative human-in-the-loop adversarial writing, producing examples that deliberately target known model shortcuts. Evaluating performance on ANLI reveals whether NLI models rely on spurious correlations rather than genuine reasoning.

In this work, we evaluate how adversarial fine-tuning on ANLI modifies the behavior of the ELECTRA-small model [6] trained on SNLI. We examine changes in overall accuracy, shifts in label prediction patterns, improvements on curated reasoning categories, and sensitivity to hypothesis and premise length. In addition, we analyze how predicted labels vary across hypothesis and premise length bins to identify and measure length-based dataset artifacts. Together, these analyses allow us to assess both robustness to adversarial examples and the degree to which ANLI fine-tuning mitigates shortcut behaviors learned from SNLI.

2 Methods

2.1 Base Model and SNLI Training

We use ELECTRA-small as our NLI model. As a baseline, we fine-tune the model only on SNLI training data. This SNLI-only model performs well on the SNLI validation (dev) dataset but performs very poorly on the ANLI dev dataset (r1+r2+r3), demonstrating limited robustness to adversarial examples.

2.2 Progressive Adversarial Fine-Tuning

We progressively fine-tune the SNLI-only trained model on the three rounds of ANLI adversarial training data [5]. For evaluation, we use the SNLI validation (dev) dataset and the ANLI dev dataset (r1+r2+r3). We evaluate the following training regimes:

1. SNLI-only + ANLI r1
2. SNLI-only + ANLI r1+r2
3. SNLI-only + ANLI r1+r2+r3
4. SNLI-only + Extra SNLI Epoch + ANLI r1+r2+r3

All models use the same hyperparameters: two epochs, batch size 32, learning rate $1e{-}5$, weight decay 0.01 (to discourage excessively large weights), and a warmup ratio of 0.1 (which gradually increases the learning rate during the first 10% of training steps to stabilize optimization).

2.3 Confusion Matrix Analysis

We compute confusion matrices for each model on ANLI dev to assess label-level changes and identify reductions in dataset artifact driven behaviors. We compare the SNLI-only model, the three ANLI fine-tuning stages (r1, r1+r2, r1+r2+r3), and the combined Extra SNLI Epoch + ANLI (r1+r2+r3) model.

2.4 Error Category Annotation

To better understand qualitative changes in model behavior, we curate a set of 20 challenging examples from ANLI dev that the SNLI-only model misclassified. Each example is manually assigned to one of five categories: paraphrase, quantity reasoning, unsupported inference, temporal reasoning, and commonsense reasoning. We then track how many errors each model corrects in each category.

2.5 Length-Based Sensitivity and Label-Bias Analysis

We group hypotheses and premises into length bins and measure error rates across both ANLI dev and SNLI dev. Because SNLI contains short, relatively simple hypotheses and premises while ANLI includes longer, multi-clause inputs, this analysis reveals whether ANLI training improves robustness to long or complex sequences. In addition to error rates, we also examine how the distribution of predicted labels varies as a function of hypothesis and premise length, enabling us to identify length-driven prediction biases present in the SNLI-trained model and to evaluate whether ANLI fine-tuning mitigates these artifacts.

3 Results

3.1 Overall Accuracy on SNLI and ANLI

Accuracy results on SNLI dev and ANLI dev are summarized in bar plots (Figures 1–2). On ANLI dev, the SNLI-only baseline performs worst (31.3%), confirming that training solely on SNLI does not prepare the model for adversarial examples. Accuracy improves steadily as additional ANLI rounds are added: 34.4% with r1, 37.5% with r1+r2, and 40.6% with r1+r2+r3, the highest overall. The Extra SNLI Epoch + ANLI combined model achieves 39.5%, slightly below the full r3 model but still far above the baseline.

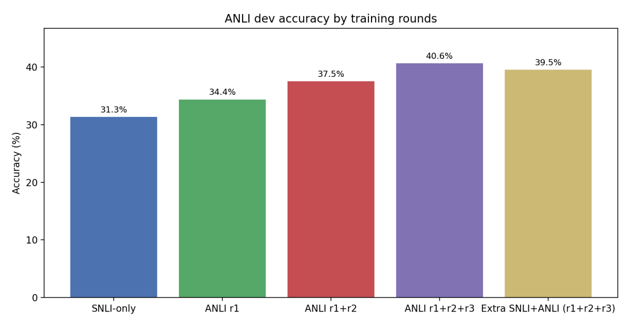

Figure 1: Dev accuracy on ANLI across training regimes.

SNLI dev accuracy remains high across all models. The SNLI-only model reaches 89.7%, while the Extra SNLI Epoch + ANLI combined model achieves 89.6%, essentially preserving in-domain performance. Other ANLI variants fall between 87.1% and 88.5%. Overall, the Extra SNLI Epoch + ANLI combined training regime offers the best balance among all settings. Specifically, it retains nearly the same SNLI dev accuracy as the SNLI-only baseline while improving ANLI dev accuracy by more than eight percentage

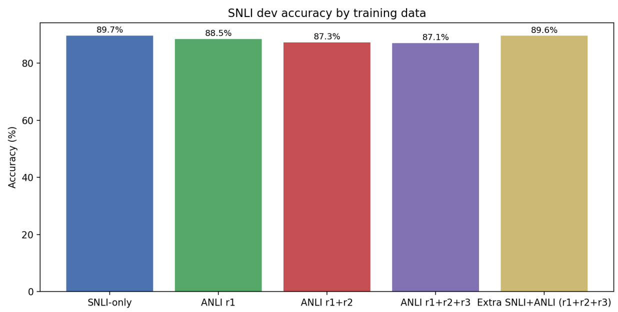

Figure 2: Dev accuracy on SNLI across training regimes.

points.

3.2 Confusion Matrix Analysis

The confusion matrices in Figures 3–7 illustrate how model behavior changes as progressively more adversarial ANLI training data is introduced. The SNLI-only model (Figure 3) exhibits a strong bias toward predicting neutral across all gold labels. For example, 61.4% of gold entailment examples and 55.8% of gold contradiction examples are incorrectly classified as neutral, showing that the baseline model relies heavily on superficial cues rather than true semantic inference.

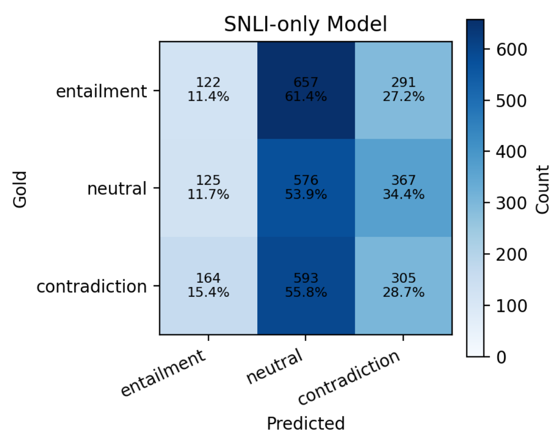

Figure 3: Confusion matrix for the SNLI-only model on ANLI dev.

Fine-tuning on ANLI produces several notable trends. First, the proportion of correctly predicted entailment examples increases dramatically from only

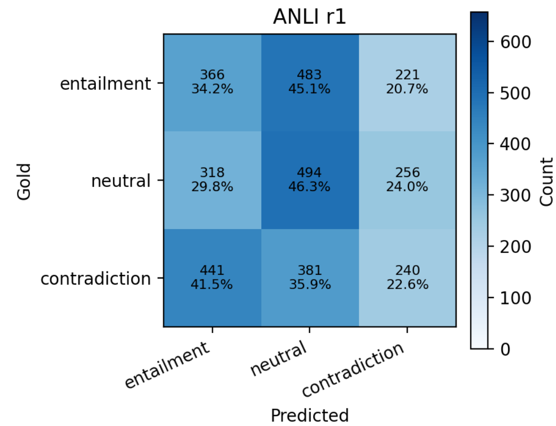

Figure 4: Confusion matrix for the ANLI r1 model on ANLI dev.

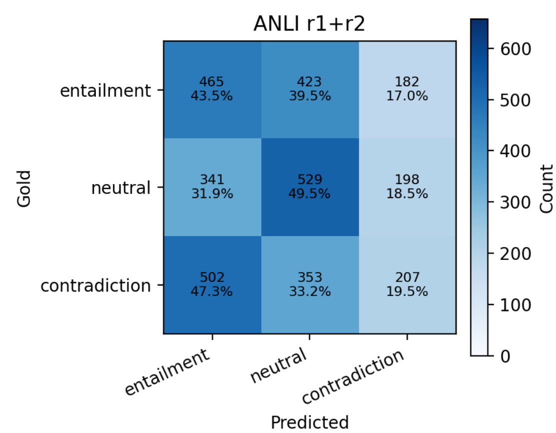

Figure 5: Confusion matrix for the ANLI r1+r2 model on ANLI dev.

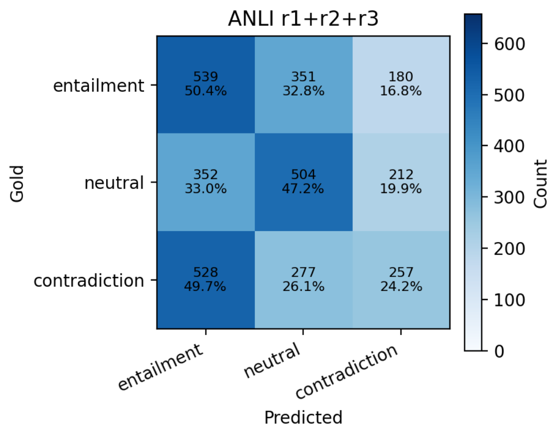

Figure 6: Confusion matrix for the ANLI r1+r2+r3 model on ANLI dev.

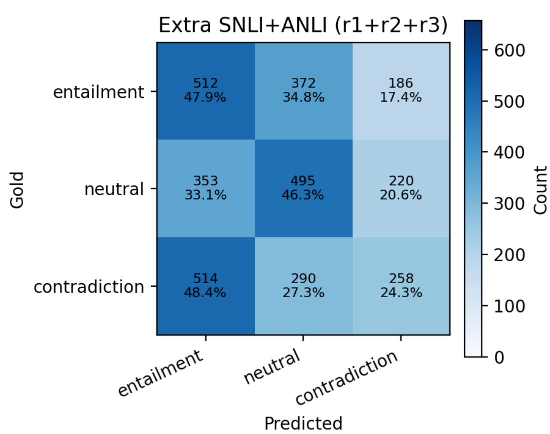

Figure 7: Confusion matrix for the Extra SNLI+ANLI (r1+r2+r3) model on ANLI dev.

11.4% in the SNLI-only baseline to as high as 50.4% in the ANLI r1+r2+r3 model. This represents the clearest improvement and indicates that adversarial training substantially enhances the model's ability to detect entailment relationships.

Second, correct neutral predictions decrease slightly relative to the baseline (53.9% down to 46.3% for r1, 49.5% for r1+r2, 47.2% for r1+r2+r3, and 46.3% for Extra SNLI Epoch + ANLI). While this appears negative at first glance, the decline is modest and actually reflects a desirable shift in that the model is no longer collapsing toward neutral as a default prediction. Instead, it is beginning to differentiate semantic relationships more carefully, rather than relying on the neutral-heavy bias inherited from SNLI.

Third, correct contradiction predictions show a mild decline (from 28.7% in the SNLI-only model to 22.6% in r1 and 19.5% in r1+r2), but partial recovery in later rounds (24.2% in r1+r2+r3 and 24.3% in Extra SNLI Epoch + ANLI). This suggests that contradiction remains difficult for ELECTRA-small even with adversarial training. The increase in incorrect contradiction to entailment predictions, from 15.4% in SNLI-only to 49.7% in the strongest ANLI models, indicates that ANLI pushes the model to consider more nuanced semantic relationships but may occasionally overshoot, interpreting strong opposing statements as logically connected.

Taken together, these trends show that ANLI fine-tuning offers substantial improvements in entailment recognition, reduces harmful overreliance on neutral, and maintains generally stable performance on contradiction, though contradiction remains the weakest class. The fact that the Extra SNLI Epoch + ANLI model nearly matches the performance of the r1+r2+r3 model suggests that once adversarial robustness is learned, it is not undone by revisiting SNLI. This indicates that ANLI induces a more durable shift in the model's internal representations, moving it away from SNLI's artifact-driven behavior and toward more semantically grounded inference.

3.3 Category-Level Improvements on Curated ANLI Errors

To better understand the types of reasoning improved by adversarial training, we evaluate all models on a curated set of 20 ANLI dev examples grouped into the five categories described in Section 2.4. The resulting counts are shown in Figure 8.

The ANLI r1 model already corrects a noticeable number of errors, particularly in quantity reasoning and temporal reasoning, but still struggles with commonsense reasoning. Adding ANLI r2 leads to further gains, especially in paraphrase, temporal reasoning, and commonsense reasoning, though some unsupported inference and quantity reasoning errors remain unresolved. With the addition of ANLI r3, the model corrects all the remaining errors. Interestingly, the Extra SNLI Epoch + ANLI combined-training model matches the performance of the full r1+r2+r3 model across all five categories, suggesting that reintroducing SNLI does not reduce adversarial robustness, at least for these examples.

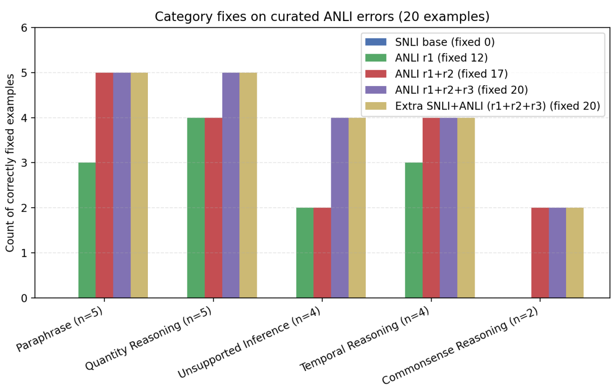

Figure 8: Number of corrected curated ANLI examples by category.

In aggregate, the strongest improvements occur in categories that require distinguishing subtle wording differences or numerical details, while more challenging categories like unsupported inference and commonsense reasoning benefit most from the later ANLI rounds.

3.4 Length-Based Error Analysis

Figures 9–12 display error rate by hypothesis and premise length and reveal how models respond to longer or more complex inputs. On ANLI dev, the SNLI-only model exhibits consistently high error rates across hypothesis length bins, typically around 65–75% for hypotheses under 30 words and rising to nearly 80% for mid-length hypotheses before dropping sharply for extremely long hypotheses (which are rare). This suggests that the baseline model is not well-equipped to handle longer or more information-dense hypotheses. In contrast, all ANLI-trained models show noticeably lower error rates across the same bins, with the full ANLI r1+r2+r3 model achieving the most stable performance and avoiding the sharp spikes present in the baseline. The Extra SNLI Epoch + ANLI combined model performs similarly well, matching r1+r2+r3 on long hypotheses and showing lower variance across bins.

A similar trend appears when analyzing premise length on ANLI dev: the SNLI-only model suffers large error spikes for mid-range premises (10–49 words), reaching over 80% error for 10–19-word premises, while all ANLI-trained models substantially reduce these peaks. The r1+r2+r3 model produces the flattest curve, demonstrating improved robustness to longer, more complex adversarial premises.

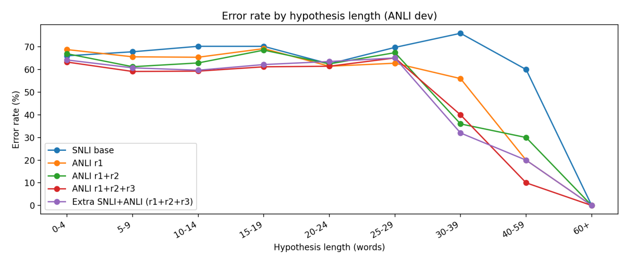

Figure 9: Error rate by hypothesis length for ANLI dev.

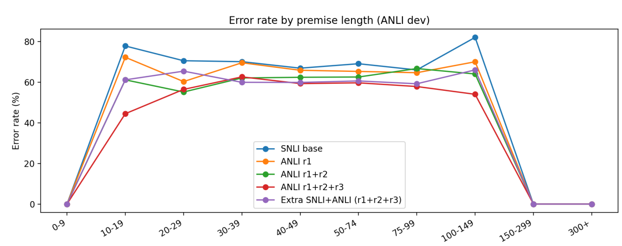

Figure 10: Error rate by premise length for ANLI dev.

On SNLI dev, all models show consistently low error rates across nearly all hypothesis and premise lengths. Error rates remain around 8–20% across most length bins, and no model shows large instability or degradation relative to the SNLI-only baseline. The only noticeable irregularities occur in bins with very few samples, where variance can spike due to the small sample size. Importantly, ANLI training does not negatively impact performance on any substantial portion of SNLI. In fact, all ANLI-trained models track closely with the SNLI-only model across the primary distribution of hypothesis and premise lengths. These patterns suggest that the SNLI-only model exploits length-based artifacts in the data rather than performing robust inference, while ANLI training helps break this dependency.

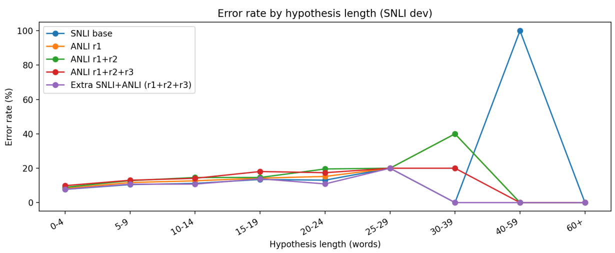

Figure 11: Error rate by hypothesis length for SNLI dev.

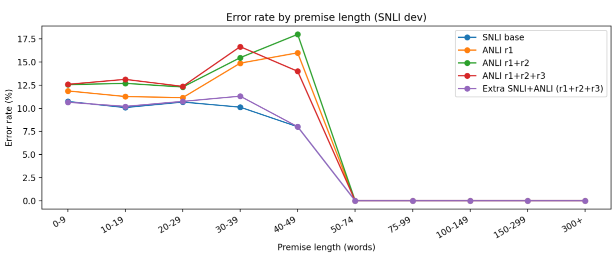

Figure 12: Error rate by premise length for SNLI dev.

3.5 Label Prediction Biases as a Function of Input Length

To further analyze length-related artifacts, we measure how each model distributes its predicted labels across bins of hypothesis and premise length (see Figures 13–14 below).

Across hypothesis-length bins, the SNLI-only model predicts neutral for the majority of inputs, regardless of length, with the exception of the 40–59 word bin where predictions become more evenly distributed across labels. This strong bias toward neutral suggests that the SNLI-only model relies on length-based shortcuts rather than true semantic inference. As ANLI rounds are added, the label distribution becomes more balanced across hypothesis lengths, indicating that adversarial training disrupts these shortcuts. Although the 40–59 word range remains dominated by neutral and contradiction even after ANLI training, the overall pattern shows a substantial reduction in length-driven biases. Notably, the Extra SNLI Epoch + ANLI model also produces label distributions that closely mirror those of the full ANLI r1+r2+r3 model, suggesting that reintroducing SNLI after adversarial training does not cause the model to revert to SNLI-style artifacts. Instead, it maintains the robustness gained from ANLI, indicating a stable shift toward meaning-driven prediction rather than reliance on superficial length cues.

Premise-length analysis shows similar but more pronounced trends. As premise length increases, the SNLI-only model overwhelmingly predicts neutral, accompanied by a sharp drop in entailment and a slight rise in contradiction. This pattern indicates that the baseline model has implicitly learned to predict neutral for longer premises, regardless of semantic content. After fine-tuning on ANLI, the models display a far more balanced label distribution across premise lengths. ANLI r1 and r1+r2 remain relatively even up through medium-length premises (0–74 words) but begin drifting toward neutral for longer spans. Introducing r3 stabilizes the distribution even in the longest bins, reducing the model's dependence on premise-length heuristics. Importantly, the Extra SNLI + ANLI model closely matches the r1+r2+r3 distribution, showing that an additional epoch of SNLI does not reintroduce SNLI-specific artifacts. Instead, it preserves the robustness gained from ANLI, suggesting that adversarial fine-tuning fundamentally shifts the model toward more semantic, rather than heuristic-driven, inference

This analysis provides direct evidence that ANLI mitigates a dataset artifact, label–length correlations, and helps the model avoid shortcut reasoning based on hypothesis and premise lengths.

3.6 Summary of Findings

Across all experiments, adversarial ANLI training improves robustness while preserving in-domain SNLI performance. Accuracy on ANLI dev increases steadily with each additional round, while SNLI dev accuracy remains nearly unchanged. Confusion matrices show that ANLI training reduces the SNLI model's strong bias toward predicting neutral and

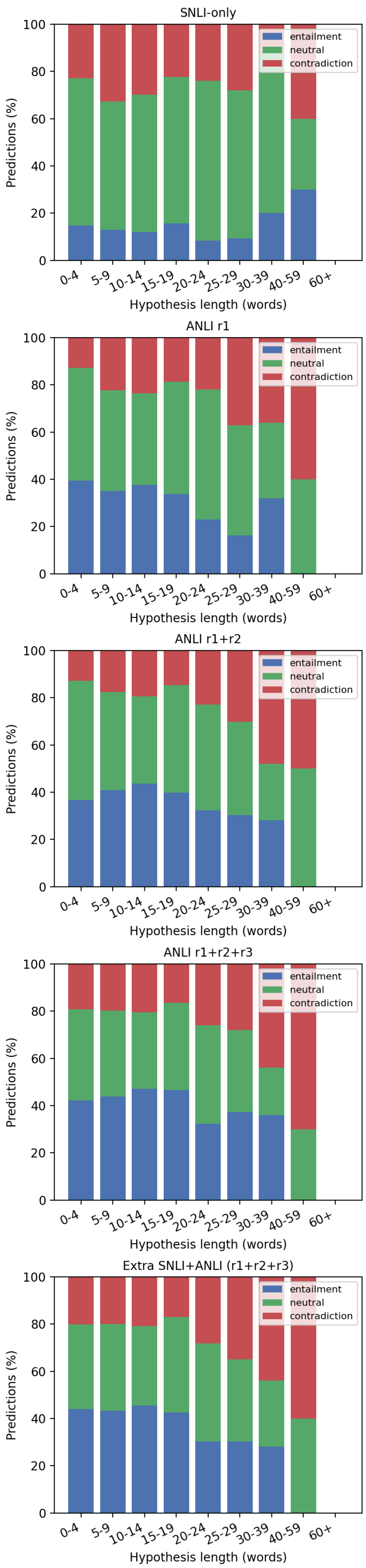

Figure 13: Distribution of predicted labels based on hypothesis length.

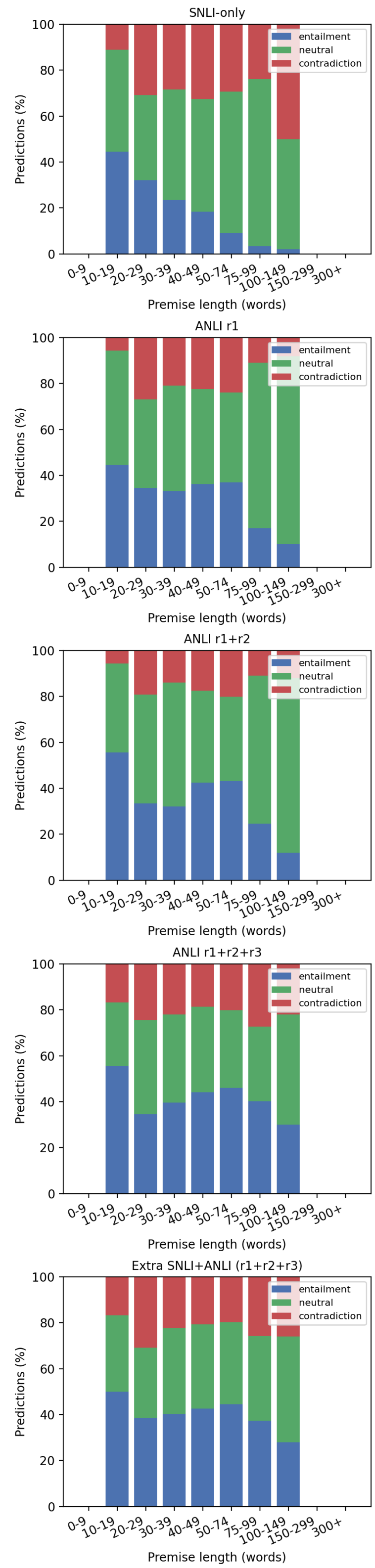

Figure 14: Distribution of predicted labels based on premise length.

vastly improves its ability to correctly distinguish entailment, but it still struggles with correctly predicting contradictions. On the curated set of challenging ANLI examples, the ANLI models progressively improve in each category. Length-based analysis demonstrates that ANLI fine-tuning improves model robustness on ANLI dev examples with long or complex hypotheses and premises, while SNLI performance remains stable across length ranges. Together, these results indicate that ANLI training teaches deeper and more robust inference patterns that are not learned from SNLI alone.

4 Conclusion

Models trained solely on SNLI rely heavily on dataset artifacts such as length-based heuristics and a strong default bias toward neutral, leading to poor performance on adversarial examples. Progressive adversarial fine-tuning with ANLI substantially improves robustness across accuracy, label prediction behavior, curated reasoning categories, and length-sensitive prediction patterns. Moreover, the Extra SNLI Epoch + ANLI model retains these improvements, demonstrating that robustness gains are stable and not easily undone by returning to non-adversarial data. Overall, these gains come with minimal cost to SNLI performance, indicating that ANLI complements rather than replaces standard SNLI supervision.

Future work may explore whether these trends extend to larger pretrained models or whether combining ANLI with additional debiasing strategies can further improve contradiction prediction. Overall, our results highlight ANLI fine-tuning as an effective and practical strategy for producing more robust and semantically grounded NLI models.

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